

Module 1 — Introduction to Matrices

Part A. Source: STPPT1. Fill this in by hand while you read.

Name _____ Date _____

Define MATRIX. The slides say a rectangular array of what, arranged how?

What does the SIZE $m \times n$ mean? Which number comes first?

Given a_{ij} , what do i and j stand for? Find the value at a position.

List every TYPE of matrix and draw a small example of each.

How is a matrix named / written? ($A = [a_{ij}]$)

SUMMARY — define a matrix, its size, and how to locate any element a_{ij} .

Module 1 — Equality, Addition, and Scalar Multiplication

Part A. Source: STPPT2, first half. The named properties get asked directly.

EQUALITY: when are two matrices equal? What must be true of their sizes?

Why is matrix equality an EQUIVALENCE RELATION?
Name its three properties.

Worked type: given $A = B$, solve for x, y, z by matching entries.

ADDITION / SUBTRACTION: what is the rule about size? What happens if it fails?

Name the FOUR properties of matrix addition and write an example of each.

SCALAR multiplication: what does k do to every entry?

SUMMARY — list the four properties of addition and the three equivalence properties.

Module 1 — Multiplication and Unary Operations

Date _____

Part A. Source: STPPT2, second half. The dimension rule is marked wrong most often.

MULTIPLICATION rule: the number of COLUMNS of A must equal what in B?

If A is $m \times n$ and B is $n \times p$, what is the size of AB?

Draw the row-times-column picture for entry (i, j) of AB.

Is AB the same as BA? Show a counter-example.

UNARY operations: what does the TRANSPOSE A^T do?

Write the properties of the transpose.

SUMMARY — state the dimension rule for AB and why order matters.

Module 2 — Elementary Row Operations

Name _____ Date _____

Part B. Source: STPDF1. This is the FA1 topic. Symbols AND words for each.

Name the **THREE** elementary row operations, in symbols and in words.

Which one has a restriction on the constant? What is it, and why?

The slides note these also work on **COLUMNS**. What changes?

Practise their notation: $2R_2 + R_1 \rightarrow R_1$ means what, exactly?

What is an **AUGMENTED** matrix? When do you use one?

Do row operations change the solution set? Why does that matter?

SUMMARY — list the three row operations from memory, without looking.

Name _____ Date _____

Date _____

Part C — Formula and Reference Sheet

Everything you should be able to recall in FA1. Keep this one beside you.

Anatomy

A matrix is a rectangular array of numbers arranged in rows and columns. An $m \times n$ matrix has m rows and n columns; $m \times n$ is its order (or dimension). The entry a_{ij} sits in row i , column j . Row index always comes first.

Matrix types

Row matrix	one row only, $1 \times n$
Column matrix	one column only, $m \times 1$
Square matrix	$m = n$
Zero / null matrix	every entry is 0
Diagonal	square, all off-diagonal entries 0
Scalar	diagonal with all diagonal entries equal
Identity I	diagonal with every diagonal entry 1; $AI = IA = A$
Upper triangular	square, all entries below the diagonal are 0
Lower triangular	square, all entries above the diagonal are 0
Symmetric	$A^T = A$

Equality of matrices

$A = B$ if and only if they have the SAME size and $a_{ij} = b_{ij}$ for every i and j . Matching entries one by one is how you solve for unknowns: set corresponding entries equal, form equations, then solve the system.

Equality is an EQUIVALENCE RELATION because it is reflexive ($A = A$), symmetric (if $A = B$ then $B = A$), and transitive (if $A = B$ and $B = C$ then $A = C$).

Operations

- Addition / subtraction: only for matrices of the SAME size. Add entry by entry. Different sizes cannot be added or subtracted, the result does not exist.
- Scalar multiplication: kA multiplies EVERY entry by k .
- Multiplication: the number of COLUMNS of A must equal the number of ROWS of B . A ($m \times n$) times B ($n \times p$) gives AB ($m \times p$).
- Entry (i, j) of AB = (row i of A) times (column j of B), paired off and summed.
- Transpose (a unary operation): A^T swaps rows and columns. $(A^T)^T = A$, $(A + B)^T = A^T + B^T$, $(AB)^T = B^T A^T$.

The four properties of matrix addition

Commutative Law	$A + B = B + A$
Associative Law	$A + (B + C) = (A + B) + C$
Identity Property	$A + 0 = 0 + A = A$, where 0 is the zero matrix of the same size
Inverse Property	$A + (-A) = 0$, where $-A$ has entries $-a_{ij}$

Expect a question that shows a worked line and asks WHICH property it illustrates. If terms were reordered it is Commutative; if only the grouping changed it is Associative.

Watch out: AB is NOT generally equal to BA . Matrix multiplication is not commutative.

Part C — Elementary Row Operations

Source: STPDF1. Learn these three in both symbols and words.

The three elementary row operations

- | | |
|-------------------------------------|---|
| 1. $R_i \leftrightarrow R_j$ | Interchange two rows. |
| 2. $kR_i \rightarrow R_i, k \neq 0$ | Multiply a row by a NONZERO scalar. |
| 3. $R_i + kR_j \rightarrow R_i$ | Add a multiple of one row to another row. |

Matrices related by these operations are ROW EQUIVALENT, written $A \sim B$. Row operations never change the solution set of the system, which is exactly why elimination is valid. The slides also state that all three correspond to ELEMENTARY COLUMN OPERATIONS: just replace the word row with column in each one.

Your module's notation

STPDF1 writes the multiplier first, e.g. $2R_2 + R_1 \rightarrow R_1$ means "2 times row 2, plus row 1, replaces row 1". Read the arrow as "becomes the new". The row named AFTER the arrow is the one that changes.

Part C — Echelon Forms, Elimination, Rank

The half of the sheet that FA1 will actually test.

Row Echelon Form (REF)

1. Any all-zero rows are at the bottom.
2. The leading entry (pivot) of each nonzero row is strictly to the RIGHT of the pivot above it.
3. All entries below a pivot are 0.

Note: some textbooks ALSO require every pivot to be a leading 1 in REF. Check which convention your prof marks against. RREF is unique for a given matrix; REF is not.

Reduced Row Echelon Form (RREF)

Everything REF requires, PLUS:

4. Every pivot is exactly 1 (a leading 1).
5. Each leading 1 is the ONLY nonzero entry in its column, so there are zeros above it too.

REF

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

RREF

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Two elimination methods

GAUSSIAN ELIMINATION reduces the augmented matrix to REF, then you finish with BACK SUBSTITUTION, solving from the bottom row upward.

GAUSS-JORDAN ELIMINATION keeps going all the way to RREF. No back substitution is needed, because each variable is already isolated and you read the answer straight off the last column.

Rank, and reading the answer

RANK = the number of nonzero rows once the matrix is in REF (or RREF). It counts the pivots.

- UNIQUE solution: rank of coefficient matrix = rank of augmented matrix = number of variables.
- INFINITELY MANY: the two ranks are equal, but LESS than the number of variables. The variables without pivots are free.
- NO SOLUTION (inconsistent): a row reads $[0 \ 0 \ \dots \ 0 \ | \ k]$ with $k \neq 0$, which says $0 = k$. The ranks differ.

Exam habit worth building

Write the operation you used next to every single row you change, e.g. $R_2 - 2R_1 \rightarrow R_2$. It earns method marks when the arithmetic slips, and it makes your own checking possible.

Part D — Practice Set

Work these on paper. Answer key is on the last page, so do not peek early.

Given

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 5 \\ -2 & 0 \end{bmatrix}$$

1. Compute $A + B$, $2A$, AB , and A^T .
2. Compute BA as well. Is $AB = BA$? State what this shows.
3. Name the type of each matrix: (a) a 1×4 matrix, (b) a square matrix whose off-diagonal entries are all 0, (c) a matrix with $A^T = A$.
4. Reduce to ROW ECHELON FORM, writing the operation beside each row you change:

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 5 & 1 \\ 3 & 7 & 2 \end{bmatrix}$$

5. Solve by GAUSS-JORDAN elimination (reduce the augmented matrix to RREF):
$$\begin{aligned} x + 2y &= 5 \\ 3x - y &= 1 \end{aligned}$$
6. Solve this 3-variable system. State the rank of the coefficient matrix and say why the solution is unique.
$$\begin{aligned} x + y + z &= 6 \\ 2x - y + z &= 3 \\ x + 2y - z &= 2 \end{aligned}$$
7. Row reduce and explain what goes wrong. What is the system called?
$$\begin{aligned} x + y &= 2 \\ 2x + 2y &= 5 \end{aligned}$$

8. MATRIX EQUALITY. Given that the two matrices are equal, find x and y .

$$\begin{bmatrix} 2x & 6 \\ 10 & y+4 \end{bmatrix} = \begin{bmatrix} 8 & 6 \\ 10 & 9 \end{bmatrix}$$

9. IDENTIFY THE PROPERTY of addition illustrated by each line:
(a) $(P + Q) + R = P + (Q + R)$
(b) $P - Q + R = P + R - Q$
(c) $P + 0 = P$
(d) $P + (-P) = 0$
10. Perform the operation in your module's notation: $3R1 + R2 \rightarrow R2$ on the matrix below. State which row changes.

$$\begin{bmatrix} 1 & -2 & 4 \\ -3 & 5 & 0 \end{bmatrix}$$

Part D — Answer Key

Check your method, not only your final number.

1 and 2

$A + B =$

$$\begin{bmatrix} 3 & 4 \\ 1 & 4 \end{bmatrix}$$

$2A =$

$$\begin{bmatrix} 4 & -2 \\ 6 & 8 \end{bmatrix}$$

$AB =$

$$\begin{bmatrix} 4 & 10 \\ -5 & 15 \end{bmatrix}$$

$A^T =$

$$\begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$$

$BA =$

$$\begin{bmatrix} 17 & 19 \\ -4 & 2 \end{bmatrix}$$

$AB \neq BA$, so matrix multiplication is NOT commutative. Order matters, and you cannot rearrange factors the way you would with real numbers.

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(a) row matrix (b) diagonal matrix (c) symmetric matrix

4 Row echelon form

$R_2 - 2R_1 \rightarrow R_2$, then $R_3 - 3R_1 \rightarrow R_3$, then $R_3 - R_2 \rightarrow R_3$.

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{bmatrix}$$

Three nonzero rows, so the rank is 3.

Check which REF convention your prof uses.

Some texts require every pivot to be a leading 1, which makes the last row $[0 \ 0 \ 1]$ instead of $[0 \ 0 \ 2]$. Both are row-equivalent and both have rank 3. RREF is unique; REF is not.

5 Gauss-Jordan

Augmented $[[1, 2 \mid 5], [3, -1 \mid 1]]$. $R_2 - 3R_1 \rightarrow R_2$ gives $[0, -7 \mid -14]$. Then $(-1/7)R_2 \rightarrow R_2$ gives $[0, 1 \mid 2]$. Then $R_1 - 2R_2 \rightarrow R_1$ gives $[1, 0 \mid 1]$.
 $x = 1, y = 2$. Check: $1 + 2(2) = 5$ ✓ and $3(1) - 2 = 1$ ✓

6 Three variables

$R_2 - 2R_1 \rightarrow R_2$ gives $[0, -3, -1 \mid -9]$. $R_3 - R_1 \rightarrow R_3$ gives $[0, 1, -2 \mid -4]$. Swap R_2 and R_3 , then $R_3 + 3R_2 \rightarrow R_3$ gives $[0, 0, -7 \mid -21]$, so $z = 3$.
Back substitute: $y - 2(3) = -4$ so $y = 2$, then $x + 2 + 3 = 6$ so $x = 1$.
Solution $(x, y, z) = (1, 2, 3)$. Rank of the coefficient matrix is 3, which equals the rank of the augmented matrix AND equals the number of variables, so the solution is unique.
Check: $1 + 2 + 3 = 6$ ✓ $2 - 2 + 3 = 3$ ✓ $1 + 4 - 3 = 2$ ✓

7 The trap

$R_2 - 2R_1 \rightarrow R_2$ gives the row $[0, 0 \mid 1]$, which reads $0x + 0y = 1$, that is $0 = 1$. That is impossible, so there is NO SOLUTION and the system is INCONSISTENT.
Rank of the coefficient matrix is 1 but rank of the augmented matrix is 2. Whenever those two ranks differ, the system is inconsistent. Geometrically the two lines are parallel and never meet.

8 Matrix equality

Equal matrices means matching entries are equal. $2x = 8$ gives $x = 4$, and $y + 4 = 9$ gives $y = 5$. The entries 6 and 10 already agree, so they give no new information.

9 Identify the property

(a) Associative Law, only the grouping changed. (b) Commutative Law, the terms were reordered.
(c) Identity Property, adding the zero matrix. (d) Inverse Property for Addition.

10 Row operation in module notation

$3R_1 + R_2 \rightarrow R_2$ changes ROW 2 only. Row 1 is used but left alone.

New $R_2 = 3(1 \ 2 \ 4) + (-3 \ 5 \ 0) = (3 \ 3 \ -6 + 5, 12 + 0) = (0, -1, 12)$.

$$\begin{bmatrix} 1 & -2 & 4 \end{bmatrix}$$